Opgaver, kapitel 3

Vektorregning

46. Vector $\vec{A}$ has magnitude 1.3 m and points 30° clockwise from the x-axis. Vector $\vec{B}$ has magnitude 1.9 m. Find the direction of $\vec{B}$ such that $\vec{A} + \vec{B}$ is in the y-direction.

Anvend vektorregning til bevægelse i 2D

57. An object is initially moving in the x-direction at 5.5 m/s, when it undergoes an acceleration in the y-direction for a period of 22 s. If the object moves equal distances in the x- and y-directions during this time, what's the magnitude of its acceleration?

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Bevægelser i 2D

73. The table below lists position
DATA versus time for an object moving in the x–y plane, which is horizontal in this case. Make a plot of position y versus x to determine the nature of the object's path. Then determine the magnitudes of the object's velocity and acceleration.

Time, t (s)	x (m)	y (m)	Time, t (s)	x (m)	y (m)
0	0	0	0.70	2.41	3.15
0.10	0.65	0.09	0.80	2.17	3.75
0.20	1.25	0.33	0.90	1.77	4.27
0.30	1.77	0.73	1.00	1.25	4.67
0.40	2.17	1.25	1.10	0.65	4.91
0.50	2.41	1.85	1.20	0.00	5.00
0.60	2.50	2.50			

80. A diver leaves a 3-m board on a trajectory that takes her 2.5 m above the board and then into the water 2.8 m horizontally from the end of the board. At what speed and angle did she leave the board?

82. You're a consulting engineer specializing in athletic facilities,
CH and you've been asked to help design the Olympic ski jump pictured in Fig. 3.27. Skiers will leave the jump at 28 m/s and 9.5° below the horizontal, and land 55 m horizontally from the end of the jump. Your job is to specify the slope of the ground so skiers' trajectories make an angle of only 3.0° with the ground on landing, ensuring their safety. What slope do you specify?

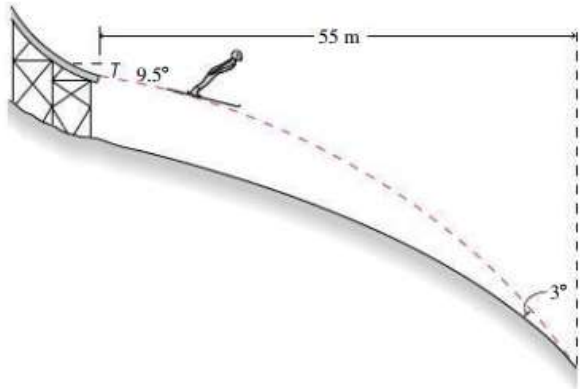

FIGURE 3.27 Problem 82

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